

to standard or existing network infrastructure. The benefit is that the system is low-maintenance with no IT integration needed and the sensors are powered by mains or long-life batteries with wireless charging option, which makes it easy to reposition them. Security is ensured by encrypting all the stored data in secure Microsoft data centres. The modular nature of the system enables easy addition and reconfiguration of new sensors,

functions, and algorithms to adapt to several needs of the users.

Besides leveraging available hospital infrastructure, the mobile app allows users to access the Vitalerter platform and is available on both Android and iOS for free. The target is to serve long-term care facilities, hospitals, and telecare providers, and improve safety, productivity, and efficiency with personalised care.

No special training is needed for

the medical staff to start using the platform for advanced analytics. The startup claims an overall 42 per cent reduction in resident falls and 68 per cent reduction in a pressure ulcer. The system has been used in over ten thousand beds in Israel. Its adoption in other countries is also widely growing since it can be used for better management by integrating and not entirely replacing human assistance with AI.

—Deepshikha Shukla

5 A SOLUTION TO FIGHT THE MOSQUITO MENACE

When we think about the deadliest animals and insects in the world, snakes, sharks, crocodiles, spiders top our list of answers. Contrary to these beliefs, mosquitoes are the vectors for diseases like malaria, Zika, dengue, chikungunya, and yellow fever that are responsible for the largest number of human deaths every year. WHO reported 438,000 deaths caused by malaria alone in 2015. On World Mosquito Day 2018, Bill Gates mentioned that mosquitoes kill more people in a day than sharks do in a century.

Out of thousands of mosquito species, the disease-causing ones were primarily confined to the tropics earlier but global warming is causing them to move into more temperate zones. Dengue, which existed in only nine countries in the 1970s, is now endemic in 128 countries and affects around 96 million people each year, according to global data collated by WHO. From employing drones to vaccines, different techniques are being used to deal with the challenge.

Understanding the seriousness of the issue, an IIT-Madras incubated startup MGH Labs developed its first simple, non-toxic, sustainable innovation, Bogorchid with the aim of shrinking mosquito population to eradicate diseases by leveraging science and technology. Weighing nearly



one kg, one Bogorchid machine effectively covers up to 186 sq. metres area (approximately 2000 sq. feet) which is sufficient to get rid of mosquitoes in a home. It has a cord length of 3.2 metres and can be powered by a 230V source.

The extract in the machine uses several artificially generated non-toxic elements such as human odour, generally used by mosquitoes to identify their food, to attract and trap these insects in a suffocation chamber, killing them naturally. The user can see and count the number of mosquitoes that get caught and killed on a daily basis.

Existing solutions like repellants and electrical traps have some drawbacks. Repellants come in many forms

but work in a similar way. Mosquito mats, papers, coils, etc burn toxic chemicals to create the smell. In other cases, the liquid evaporated from a bottle creates deadly fumes to deal with the mosquitoes, which affect human health too. Electric traps use ultraviolet light to lure the mosquitoes and kill every insect that comes near the electric rod. The rod gets covered with insects after a period of time and starts emitting a foul odour. Mosquito bats and nets have their limitations, too.

Unlike these products, Bogorchid provides safety from mosquito-borne diseases with eighty per cent

efficiency, while ensuring that the surrounding air is free from germs and bacteria. It is designed in India keeping in mind the country's weather conditions. The chemical emits no smell, contains no harmful contents, and has no side effects. This makes it suitable for homes with children, elders, asthma, and sinus patients.

The original model was introduced in two colours at a price of ₹ 1887 per unit. The outdoor device comes with a fine mesh catch bag and can cover a maximum range of thirty metres. Separately, the refill bottles containing synthetic pheromone to mimic odour were priced at ₹ 150 per unit. Currently, the model is being revamped and not available for purchase. **EFY**

—Ayushee Sharma